

MEENAKSHI KRISHNAN

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EDUCATION

PhD in Applied Mathematics

Fall 2021 - Present

University of Maryland, College Park.

PhD Candidate in the Applied Mathematics, Statistics, & Scientific Computation (AMSC) program.

Advisor: [Prof. Ramani Duraiswami](#).

GPA: 4.0/4.0

Bachelor and Master of Science (BS-MS) in Mathematics,

2016-2021

Indian Institute of Science Education and Research Thiruvananthapuram (IISER-TVM).

GPA: 9.56/10

RESEARCH INTERESTS

Partial Differential Equations, Machine Learning, Numerical Analysis, Acoustics, Differential Geometry, and Optimization.

PUBLICATIONS

Arun, K. R., **M. Krishnan***, and S. Samantaray. *A Unified Asymptotic Preserving and Well-Balanced Scheme for the Euler System with Multiscale Relaxation*. Computers & Fluids 233: 105248, 2022.

M. Krishnan, L. Fowl, and R. Duraiswami. *3D Gaussian Splatting with Normal Information for Mesh Extraction and Improved Rendering*. 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) Workshops. IEEE, 2025.

M. Krishnan, R. Duraiswami. *Generalization Analysis and Improved Shape Representation with Neural Signed Distance Functions*. International Conference on Learning Representations (ICLR) 2026 Workshop on AI&PDE, 2026. Spotlight Presentation (Oral).

M. Krishnan, R. Duraiswami. *ViscoReg: Neural Signed Distance Functions via Viscosity Solutions*. Transactions on Machine Learning Research (TMLR), 2026.

K. Chen, **M. Krishnan***, and H. Yang. *Error Analysis for Learning Time-Stepping Algorithms for PDEs* (in review). arXiv preprint arXiv:2509.04256, 2025.

M. Krishnan, P. Pulijala, K. Chen, H. Yang, and R. Duraiswami. *GAIA: Geometry Adaptive Integral AutoEncoder Networks for Operator Learning* (in review). arXiv preprint arXiv:2607.01128, 2026.

V. Patil, S. Patil, **M. Krishnan**, A. Kundu, C. Fermuller, R. Duraiswami, and J. JaJa. *Disentanglement via Conformal Maps for Combinatorial Generalization* (in review). 2026.

*: Primary author, alphabetical ordering.

EXPERIENCE

PhD Research Intern

May 2026 - August 2026

Audio AI, Advanced Technology Group, Dolby Laboratories

San Francisco, CA

- Developed operator learning models for audio simulations.

PhD Research Intern

Audio, Advanced Technology Group, Dolby Laboratories

May 2024 - August 2024

San Francisco, CA

- Developed neural proxies for room acoustics simulations to approximate PDE-based acoustic solvers.
- Implemented multi-physics FEM-BEM models (using COMSOL, MATLAB) for coupled Helmholtz-electromagnetic equations to train and validate deep learning models.

Graduate Teaching Assistant

Department of Mathematics, University of Maryland

Sept 2021 - June 2023

College Park, MD

- Courses: MATH120 (Elementary Calculus), MATH141 (Calculus II), MATH401 (Applications of Linear Algebra), MATH416 (Applied Harmonic Analysis), AMSC420 (Mathematical Modeling), AMSC661 (Scientific Computing II)

DAAD-WISE Research Intern

Advisor: Prof. Dr. Mária Lukáčová-Medvidová, University of Mainz

May 2019 - July 2019

Mainz, Germany

Project: Implicit Explicit Runge Kutta (IMEX-RK) schemes for Stiff Differential Equations.

Indian Academy of Science's Summer Research Fellowship

Advisor: Prof. K Sandeep, TIFR-CAM

May 2018 - July 2018

Bengaluru, India

FELLOWSHIPS

Wylie Dissertation Fellowship 2026. Award by the Graduate School at UMD.

Mark E. Lachtman Award 2026. Award by the Dept. of Mathematics at UMD.

Outstanding Research Assistant Award 2026. Award by the Graduate School at UMD.

NeuroPAC Fellowship 2025. Award by NeuroPAC and NSF for accelerating research on Neuromorphic Perception.

Hauptmann Fellowship 2024. Awarded by the Dept. of Mathematics at UMD.

Seymour Goldberg Spotlight Talks on Graduate Research Award 2024.

Janine Tucker and Ira Schwartz Endowed Graduate Award in Mathematics 2023. Award recognizing early-stage graduate students demonstrating a high level of excellence in mathematics.

Dean's Fellowship 2021-2023. Awarded by UMD Graduate School.

ANU Future Research Talent Award 2020 (canceled due to COVID19 pandemic). Award granted by the Australian National University (ANU).

DAAD-WISE Fellowship 2019. The German Academic Exchange Service's Working Internships in Science and Engineering.

Innovation in Science Pursuit for Inspired Research (INSPIRE) Fellowship 2016-2021. Awarded by the Department of Science and Technology of the Govt. of India.

LEADERSHIP

- **Treasurer, GSCAMS 2024-25.** Member of the Graduate Student Committee of AMSC, MATH, & STAT (GSCAMS) at UMD.
- **President, Women in Math (WIM) 2022-23, 2023-24.** President of UMD WIM Association.